The Complexity of Graph Isomorphism

From Factorial Brute Force to Quasi-Polynomial Breakthrough

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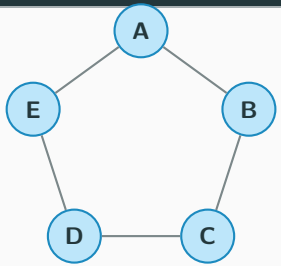
Department of Computer Science

Outline

1. Introduction — What is Graph Isomorphism?
2. Complexity — Where Does GI Live?
3. Why Computers Struggle — The Matrix Problem
4. Babai's Breakthrough — 2015
5. The Weisfeiler-Lehman Test — Solving GI in Practice
6. Algorithms — A Progression
7. Special Cases — When GI Is Easy
8. Real-World Applications

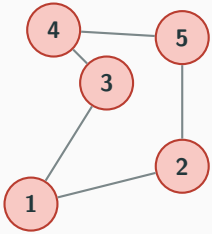
Introduction — What is Graph Isomorphism?

Are These the Same Graph?



Graph G

?



Graph H

★ Human Intuition Fails Here

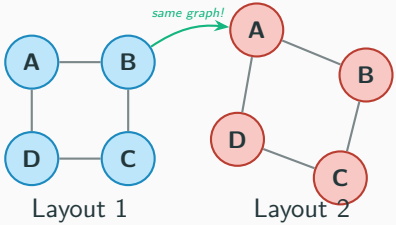
They *look* different — but are they structurally identical?
This is the **Graph Isomorphism Problem**.

The Word Isomorphism

Etymology

- **iso** (Greek) *equal*
- **morphe** (Greek) *shape*

⇒ **"Same shape"**



The Necklace Analogy

Tangle, stretch, or rearrange a necklace — the *connections* never change. The **structure** is what matters, not the picture.

Formal Definition

Definition — Graph Isomorphism

Two graphs $G = (V_G, E_G)$ and $H = (V_H, E_H)$ are **isomorphic** ($G \cong H$) if there exists a bijection:

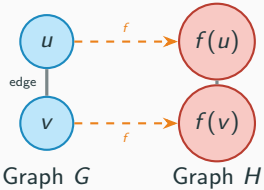
$$f : V_G \longrightarrow V_H$$

such that for all $u, v \in V_G$:

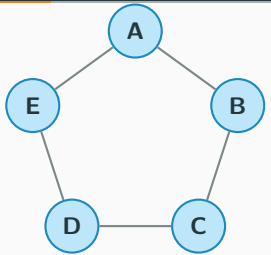
$$(u, v) \in E_G \iff (f(u), f(v)) \in E_H$$

In plain English:

- If A and B are friends in G, their twins $f(A)$, $f(B)$ must be friends in H.
- No friendship is added or removed.
- It is a *structure-preserving* renaming.

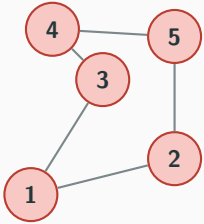


Revealing the Mapping



G

$G \to H$
A $\to$ 1
B $\to$ 2
C $\to$ 5
D $\to$ 4
E $\to$ 3



H

$G \cong H$ — **They are the same graph!**

The mapping f is the proof.

Why is This a Problem?

Brute-Force Approach:

- Try *every possible* mapping f
- For n nodes: $n!$ possibilities

Scale of $n!$

Nodes	$n!$
5	120
10	3,628,800
20	$\approx 2.4 \times 10^{18}$
100	9.33×10^{157}

$\otimes$ Atoms in the observable universe:

$$\approx 10^{80}$$

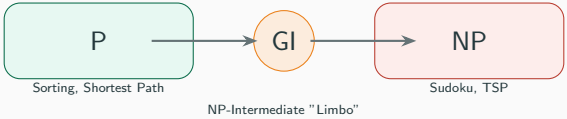
$100! = 9.33 \times 10^{157}$

Vastly exceeds the number of atoms!

$\Rightarrow$ We need a **clever algorithm**.

Complexity — Where Does GI Live?

The "Map" of Computational Complexity



Why is GI Not Known to be in P?

? Why Not Easy (P)?

- No polynomial algorithm found that works for *every* graph.
- Highly **symmetric** graphs defeat most shortcuts.
- After decades of effort — still open!

? Why Not Hard (NP-Complete)?

- NP-Complete problems have *random* structure.
- GI has rich **mathematical structure** (Group Theory).
- It's more like a Rubik's Cube than a pile of tangled yarn!

⊞ The Rubik's Cube Analogy

NP-Complete: Tangled yarn — no structure to exploit.

GI: A Rubik's Cube — looks messy, but strict mathematical rules allow clever "twists."

Evidence: GI Cannot Be NP-Complete

▷ Schöning's Theorem (1988)

If **Graph Isomorphism** were NP-Complete,
then the **Polynomial Hierarchy** would collapse.

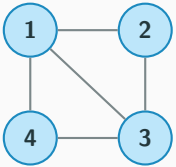
What is the Polynomial Hierarchy?

- A tower of complexity levels above P and NP.
- Each level is "strictly harder" than the one below.
- A *collapse* means many "hard" problems
  become "easy" — widely believed to be **false**.

$\Rightarrow$ GI is **almost certainly not NP-Complete**.
It lives in the mysterious **NP-Intermediate** zone.

Why Computers Struggle — The Matrix Problem

How a Computer "Sees" a Graph



Graph G

Adjacency Matrix of G:

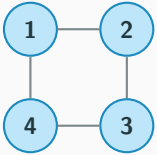
$$A_G = \begin{pmatrix} 0 & \mathbf{1} & \mathbf{1} & \mathbf{1} \\ \mathbf{1} & 0 & \mathbf{1} & 0 \\ \mathbf{1} & \mathbf{1} & 0 & \mathbf{1} \\ \mathbf{1} & 0 & \mathbf{1} & 0 \end{pmatrix}$$

- Entry $[i, j] = 1$ iff edge (i, j) exists.
- Node **labels** determine the matrix layout.

The Labelling Nightmare

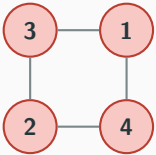
Key insight: The same graph gives *different* matrices for different labellings!

Labelling 1



$$\begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix}$$

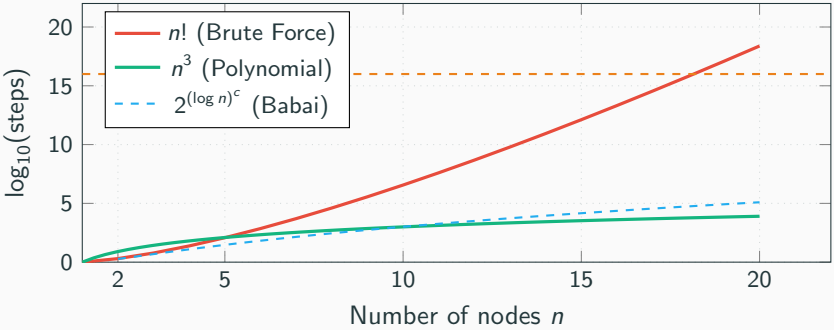
Labelling 2 (same graph!)



$$\begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix}$$

The computer must find: can we **permute rows & columns** of A_G to get A_H?
For n nodes $\Rightarrow$ $n!$ row/column permutations to check.

The Factorial Explosion



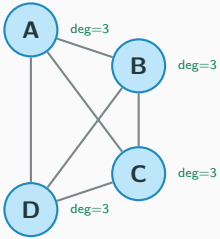
For $n = 100$: brute force needs $\sim 10^{157}$ steps — utterly infeasible.

The "No Corners" Problem — Regular Graphs

Regular Graph

Every node has the **same degree** (same number of neighbours).

- In a jigsaw puzzle ⇒ look for *corner pieces*.
- In GI with regular graphs ⇒ **all nodes look identical**.
- No unique starting point!
- The computer gets completely lost.



K_4: every node has degree 3

Babai's Breakthrough — 2015

November 2015 — A Bombshell

★ The Announcement

László Babai (University of Chicago) announced a

Quasi-Polynomial

time algorithm for Graph Isomorphism.

- Previous best: Luks (1983) — sub-exponential.
- Babai improved it by an *exponential factor*.
- Considered one of the greatest TCS results of the 21st century.

László Babai
Univ. of Chicago
Computer Scientist

What is Quasi-Polynomial Time?

Babai's Algorithm Runs in Time:

$$2^{O((\log n)^c)} \quad \text{for some constant } c$$

Exponential (Old)

2^n

For $n = 100$:

$\approx 10^{30}$ steps

Quasi-Poly (Babai)

$2^{(\log n)^c}$

For $n = 100$:

$\approx 10^7$ steps

Polynomial (Goal)

n^k

For $n = 100$:

$\approx 10^6$ steps

Exponential $\gg$ Quasi-Polynomial $\gg$ Polynomial

Babai's result is the *closest* GI has ever been to the "Easy" bucket.

Explore More: Visualize GI Complexity

Explore More

For an interactive visualization of Graph Isomorphism complexity (including Babai's breakthrough), visit:
https://graph-isomorphism-complexity.netlify.app/

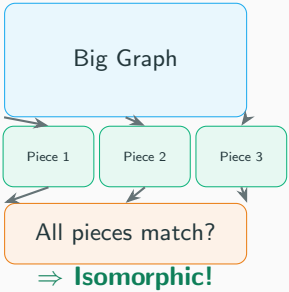
Babai's Strategy: Divide and Conquer

Three Key Ideas:

1. **Symmetry Breaking**
   Hard graphs are too symmetrical.
   Find "weak" spots to break symmetry.

2. **Local Landmarks (Splitter)**
   Identify unique sub-structures within the graph.

3. **Divide & Conquer**
   Split graph into smaller pieces at landmarks.
   If all pieces match $\Rightarrow$ whole graphs match.



Uses deep results from **Group Theory**.

The Drama Behind the Proof

△ The Gap (Jan 2017)

Harald Helfgott found a *gap* (error) in one of Babai's key lemmas.

The community held its breath...

✓ The Fix (Feb 2017)

Babai returned to his office, worked for weeks, and **fixed the proof**.

By early 2017: community consensus — *the proof is correct*.

What Babai's Result Means

▲ Impact

- **Algorithmically:** Vastly faster for large graphs.
- **Theoretically:** GI is "almost" polynomial — the closest it has ever been to P.
- **Structurally:** Confirms GI has deep mathematical structure exploitable by algorithms.

"Babai's result didn't just give us a faster tool; it proved that Graph Isomorphism is fundamentally **not that hard**.*"*

Open Question (Still!)

Is GI in **P**?
No one knows. It remains one of the great open problems in TCS.

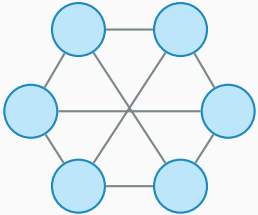
The Weisfeiler-Lehman Test — Solving GI in Practice

Real-World Solution: The Colour Refinement Trick

Weisfeiler-Lehman (WL) Test

Instead of checking structure directly, we **colour** nodes based on their neighbourhood.

If two graphs produce different colour palettes $\Rightarrow$ **not isomorphic**.

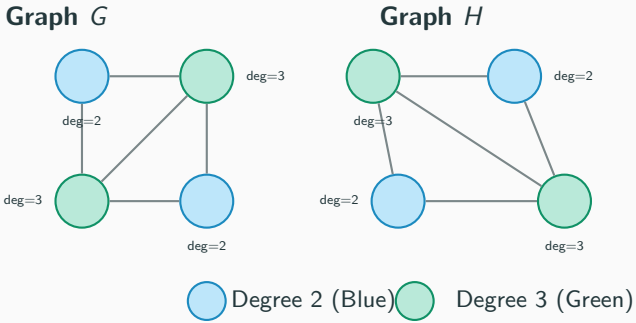


Petersen graph

Two Phases:

1. **Initial Colouring** — by node degree.
2. **Refinement** — update colour based on neighbours' colours.

WL Step 1: Initial Colouring by Degree



Both graphs have **2 blue** (degree-2) and **2 green** (degree-3) nodes.
$\Rightarrow$ Initial colouring does *not* distinguish them. **Refine further!**

WL Step 2: Refinement by Neighbourhood Colours

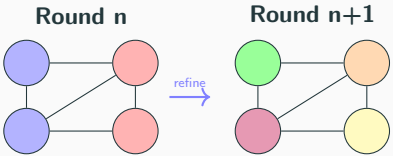
Rule: Update each node's colour based on *its own colour + sorted colours of neighbours*.

Example Update

Node **Blue** with neighbours
{**Red**, **Red**} ⇒ New color: **Purple**

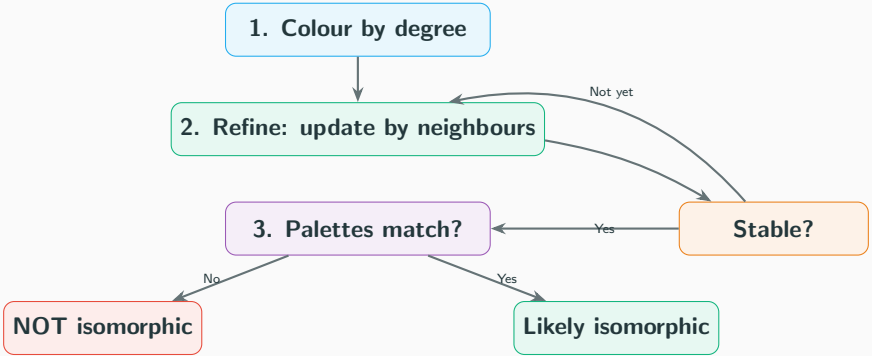
Node **Blue** with neighbours
{**Red**, **Blue**} ⇒ New color: **Green**

Repeat until colours **stabilise**[cite: 403].



Nodes change shade because their neighbors look different!

WL Algorithm — Full Flow



WL Test: Power and Weakness

(+) Strengths

- Runs in **polynomial time** $O(n \log n)$.
- Works for almost all graphs encountered in practice.
- Used in **Graph Neural Networks** (GNNs) in AI!

(-) Weakness

- Fails on **Cai-Fürer-Immerman (CFI) graphs**.
- Highly regular graphs with no distinguishing local structure.
- WL says "same" but they are *different*!

CFI Graph — The WL Adversary

WL cannot tell these apart!

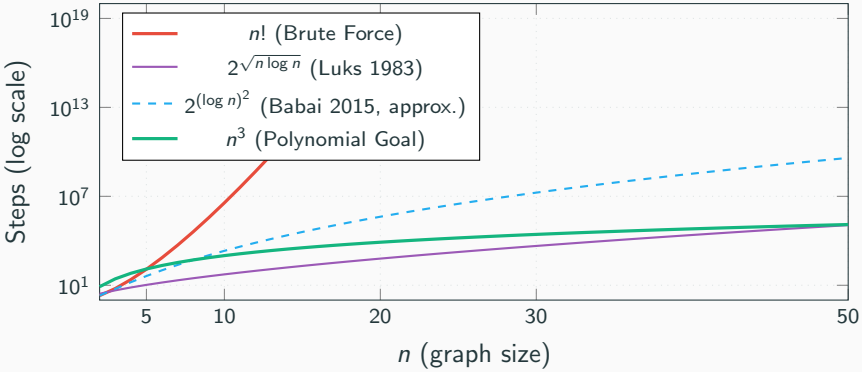
Algorithms — A Progression

Evolution of GI Algorithms

Year	Author	Complexity	Key Idea
1970s	Various	$O(n!)$	Brute force / backtracking
1980	Luks	$2^{O(\sqrt{n \log n})}$	Group theory, bounded degree
1983	Luks	Sub-exponential	Divide & conquer on groups
1992	Babai, Luks	$e^{O(\sqrt{n \log n})}$	Canonical form
2015	**Babai**	$\mathbf{2^{O((\log n)^c)}}$	**Group theory + local cert.**

Each step is a *massive* improvement. The 2015 jump was the most dramatic in 30 years.

Growth Rate Comparison



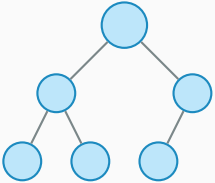
Special Cases — When GI Is Easy

When is GI Provably Easy?

✓ Polynomial Cases

- **Trees** — $O(n \log n)$
- **Planar graphs** — $O(n)$ (Hopcroft & Wong 1974)
- **Bounded degree** — polynomial (Luks 1980)
- **Interval graphs**
- **Bounded genus graphs**

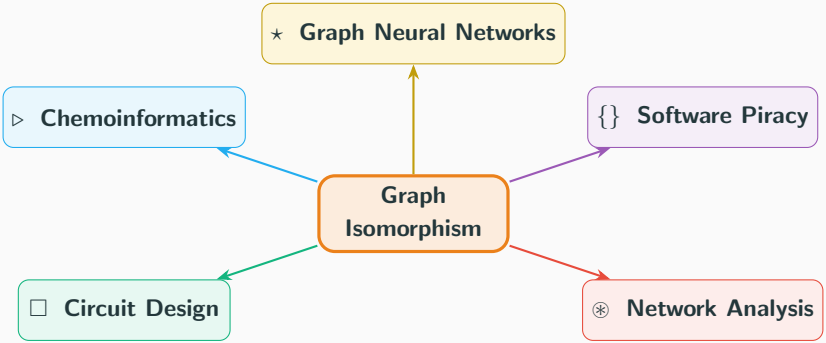
Tree (Easy!)



The more **structure** a graph family has, the easier GI becomes.
Random, highly-connected graphs are the hardest cases.

Real-World Applications

Why Does This Matter?

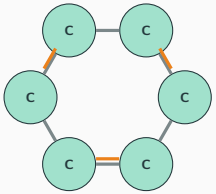


Application 1: Chemoinformatics

▷ The Problem

Are two molecular structures the same compound?

Atoms = nodes. Chemical bonds = edges.



Benzene ring as a graph

Why it matters:

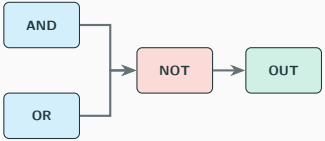
- Drug discovery: is a new compound the same as a known drug?
- Toxicology: detecting toxic isomers.
- Patent checking for novel compounds.

Application 2: Circuit Design Verification

The Problem

Does the physical chip layout match the digital blueprint?

Logic gates = nodes. Wires = edges.



Why it matters:

- Modern chips have *billions* of transistors.
- Even one wrong connection $\Rightarrow$ chip failure.
- GI algorithms enable automated verification.

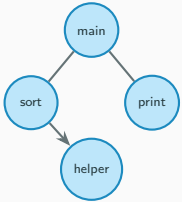
Application 3: Software Piracy Detection

{} The Problem

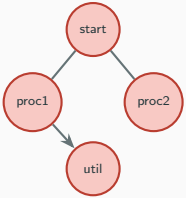
Is this code a stolen copy, even if variable names were changed?

Functions = nodes. Call relationships = edges.

Original Code (Call Graph):



Stolen Code (Renamed):



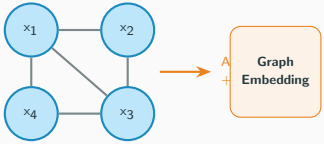
GI detects: same call structure ⇒ likely **copied code**!

Application 4: Graph Neural Networks (AI)

⋆ GNNs and WL

Modern AI uses graphs to represent: molecules, social networks, knowledge bases.

- GNNs work by *aggregating* node features from neighbours.
- This is **exactly** the WL colour refinement algorithm!
- The **expressiveness** of a GNN is bounded by the WL test.
- Key result: standard GNNs cannot distinguish non-isomorphic graphs that fool WL.



WL power

= GNN power.

Open Problems and Summary

What Remains Unknown?

1. **Is GI in P?**
   The million-dollar question. We have quasi-polynomial; polynomial is still open.
2. **Can Babai's algorithm be improved to polynomial?**
   Would require fundamentally new group-theoretic machinery.
3. **Does $P \neq NP$?**
   If $P = NP$, then GI $\in$ P trivially.
4. **Is there a graph canonical form computable in polynomial time?**
   A canonical form would solve GI; its complexity remains open.

Graph Isomorphism sits at the **frontier** of complexity theory.
It has driven some of the deepest results in algorithms and group theory.

The Full Journey of GI

Summary — The Big Picture

What is GI?

Determining if two graphs share the same structure via a bijection $f : V_G \to V_H$.

Where does it live?

NP-Intermediate — harder than P, almost certainly easier than NP-Complete.

Best algorithm?

Babai (2015): $2^{O((\log n)^c)}$ — quasi-polynomial.

Practical solution?

Weisfeiler-Lehman colour refinement — polynomial, works on almost all real graphs.

Open question?

Is GI $\in$ P? Nobody knows. Still an active frontier.

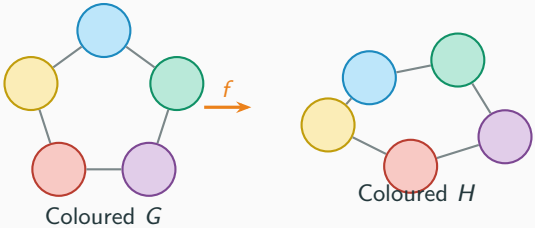
Real-world uses?

Chemistry, circuits, code, AI/GNNs, networks.

The Key Takeaway

"In theory, Graph Isomorphism is a monster.

In practice, we tame the monster with colour refinement."



Thank You!

Questions?

☒ [your.email@university.edu]

[gh] github.com/yourprofile

Appendix: Q&A Cheat Sheet

Q: Is GI the same as Graph Automorphism?

A: Related but different. Automorphism is GI with $G = H$ — finding self-symmetries.

Q: How is GI used in machine learning?

A: GNNs use WL-style aggregation. Their expressiveness is provably limited by WL.

Q: Did Babai win a prize for his result?

A: He won the Knuth Prize (2015) and it is widely regarded as a landmark result.

Q: What would happen if GI were shown to be in P?

A: It would be a major breakthrough; many subgraph problems would become faster.

Q: Can quantum computers solve GI efficiently?

A: No known quantum algorithm gives a superpolynomial speedup for GI yet.

Appendix: Key Notation Reference

Symbol	Meaning
$G = (V, E)$	Graph with vertex set V, edge set E
$G \cong H$	G and H are isomorphic
$f : V_G \to V_H$	Bijection (the isomorphism mapping)
n	Number of vertices ($\lvert V \rvert$)
$n!$	Factorial — number of brute-force mappings
$O(n^k)$	Polynomial time
$2^{O((\log n)^c)}$	Quasi-polynomial time (Babai)
$\deg(v)$	Degree of vertex v
WL	Weisfeiler-Lehman test